Caffeinated and alcoholic beverage intake in relation to ovulatory disorder infertility

Background Many studies have examined whether caffeine, alcohol, or specific beverages containing these affect fertility in women. However most of these studies have retrospectively collected information on alcohol and caffeine intake, making the results susceptible to biases. Methods We followed 18,555 married women without a history of infertility for 8 years as they attempted to become (or became) pregnant. Diet was measured twice during this period and prospectively related to the incidence of ovulatory disorder infertility. Results There were 438 incident report of ovulatory disorder infertility during follow-up. Intakes of alcohol and caffeine were unrelated to the risk of ovulatory disorder infertility. The multivariate-adjusted relative risk (RR), 95% confidence interval (CI), P for trend comparing the highest to lowest categories of intake were 1.11 (0.76–1.64; 0.78) for alcohol and 0.86 (0.61–1.20; 0.44) for total caffeine. However, intake of caffeinated soft drinks was positively related to ovulatory disorder infertility. The multivariate-adjusted RR 95% CI, and P for trend comparing the highest to lowest categories of caffeinated soft drink consumption were 1.47 (1.09–1.98; 0.01). Similar associations were observed for noncaffeinated, sugared, diet and total soft drinks. Conclusions Our findings do not support the hypothesis that alcohol and caffeine impair ovulation to the point of decreasing fertility. The association between soft drinks and ovulatory disorder infertility appears not to be attributable to their caffeine or sugar content, and deserves further investigation.